

Notes

Assignments

- 4.5
- 4.21 (b) (g) (h)
- 4.22 (c) (e)
- 4.27

Tutorial problems

- Basic Problems with Answers 4.8, 4.9
- Basic Problems 4.23
- Advanced Problems 4.39, 4.40

DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} \quad (\text{Analysis equation})$$

Different
from CT
Fourier
series



$\sum_{k=\langle N \rangle}$ = Sum over *any* N consecutive values of k

$$x[n] = x[n + N]$$

$$a_{k+N} = a_k$$

DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} \quad (\text{Analysis equation})$$

CT Fourier Series Pairs

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} = \sum_{k=-\infty}^{+\infty} a_k e^{j2\pi kt/T}$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_o t} dt$$

LTI System, system function and frequency response

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} H(jk\omega_0) a_k e^{jk\omega_0 t}$$

$$a_k \longrightarrow \underbrace{H(jk\omega_0) a_k}_{\text{"gain"}}$$

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t) e^{-j\omega t} dt$$

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 n} \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} H(e^{jk\omega_0}) a_k e^{jk\omega_0 n}$$

$$a_k \longrightarrow \underbrace{H(e^{jk\omega_0}) a_k}_{\text{"gain"}}$$

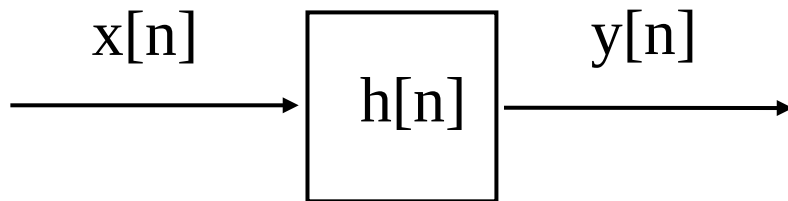
$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n] e^{-j\omega n}$$

The effect of the LTI system is to modify each a_k through multiplication by the value of the frequency response at $k\omega_0$.

Example 3.17

$$h[n] = \alpha^n u[n] \quad , \quad |\alpha| < 1$$

$$x[n] = \cos\left(\frac{2\pi n}{N}\right) = \frac{1}{2} e^{j(\frac{2\pi}{N})n} + \frac{1}{2} e^{-j(\frac{2\pi}{N})n}$$



$$H(e^{j\omega}) = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \frac{1}{1 - \alpha e^{-j\omega}}$$

$$y[n] = \frac{1}{2} H\left(e^{j\frac{2\pi}{N}}\right) e^{j(\frac{2\pi}{N})n} + \frac{1}{2} H\left(e^{-j\frac{2\pi}{N}}\right) e^{-j(\frac{2\pi}{N})n}$$

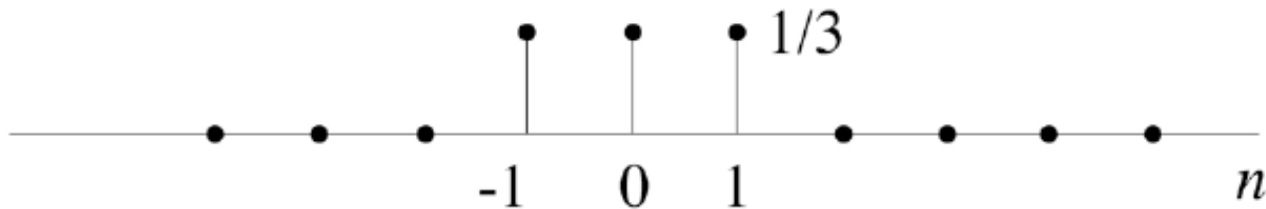
$$= r \cos\left(\frac{2\pi n}{N} + \theta\right)$$

$$\text{where } re^{j\theta} = \frac{1}{1 - \alpha e^{-j\frac{2\pi}{N}}}$$

Example #3: DT Averager/Smother

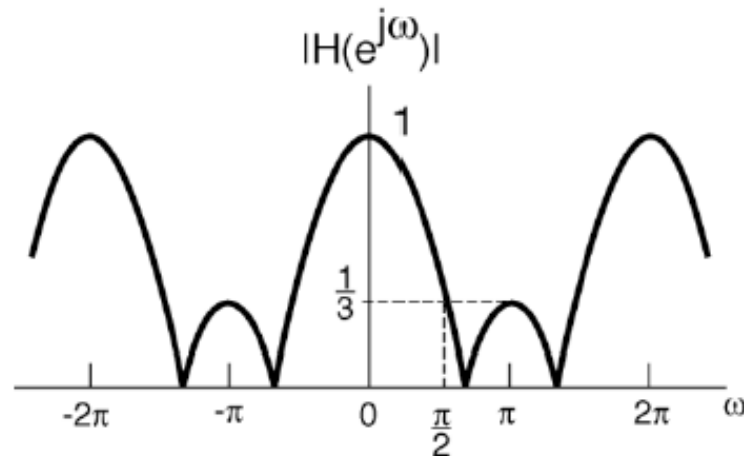
$$y[n] = \frac{1}{3}\{x[n-1] + x[n] + x[n+1]\}$$

$$h[n] = \frac{1}{3}\{\delta[n-1] + \delta[n] + \delta[n+1]\}$$



Frequency response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \frac{1}{3}[e^{-j\omega} + 1 + e^{j\omega}] = \frac{1}{3} + \frac{2}{3}\cos\omega$$



A LPF

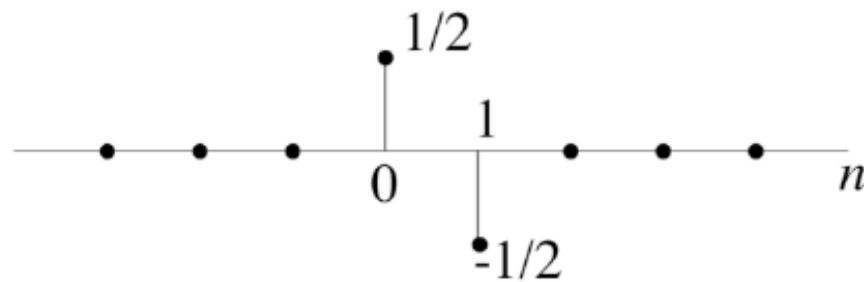
Example #5:

Simple DT “Edge” Detector

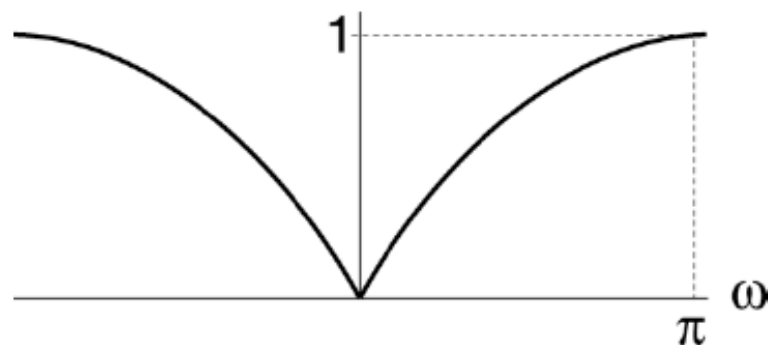
— DT 2-points “differentiator”

$$y[n] = \frac{1}{2}[x[n] - x[n-1]]$$

$$h[n] = \frac{1}{2}[\delta[n] - \delta[n-1]]$$



$|H(e^{j\omega})|$



Amplifies high-frequency components

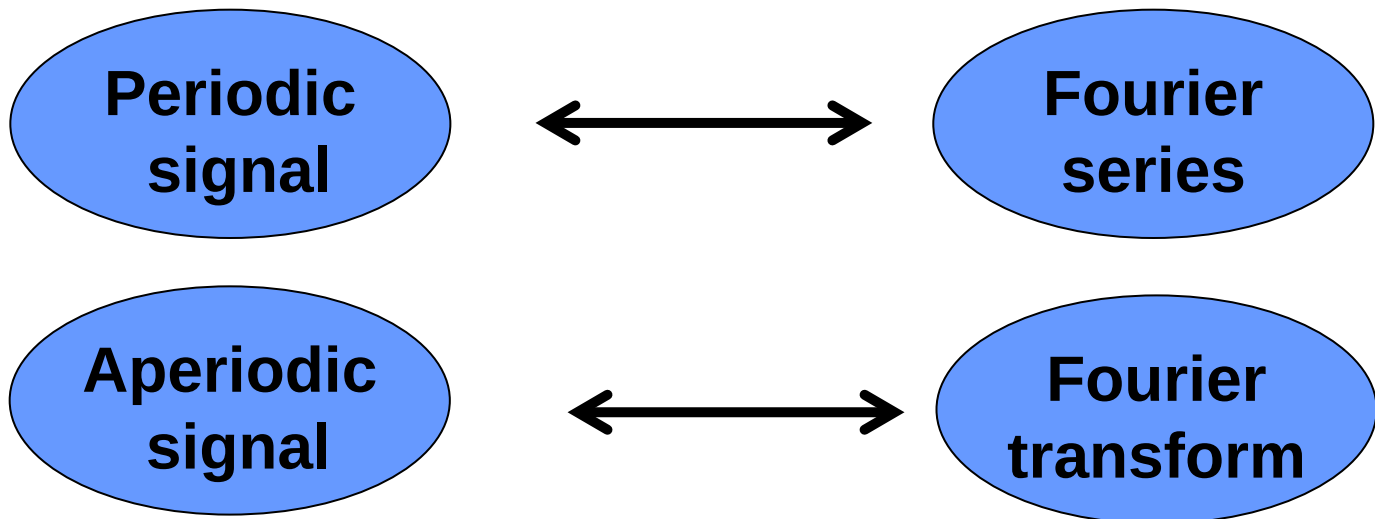
Frequency response:

$$H(e^{j\omega}) = \frac{1}{2}(1 - e^{-j\omega}) = je^{j\omega/2} \sin(\omega / 2)$$

$$|H(e^{j\omega})| = |\sin(\omega / 2)|$$

Chapter 4

The Continuous-Time Fourier Transform

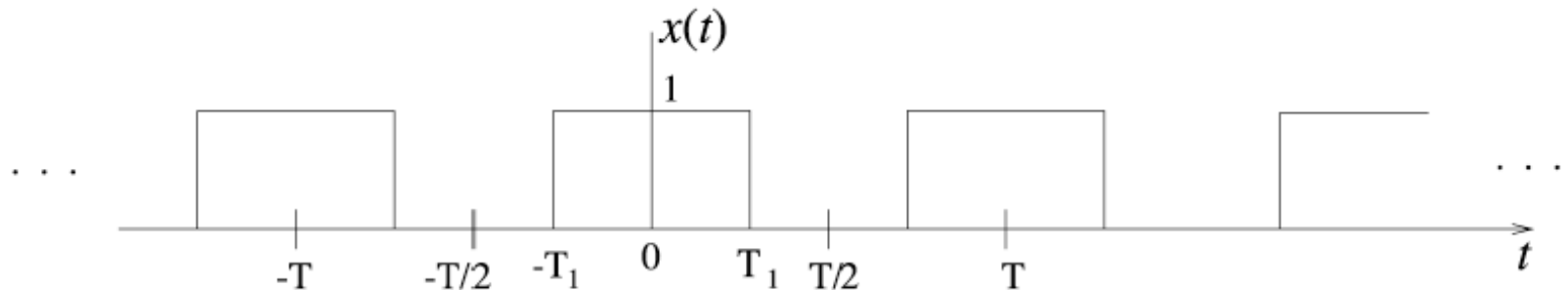


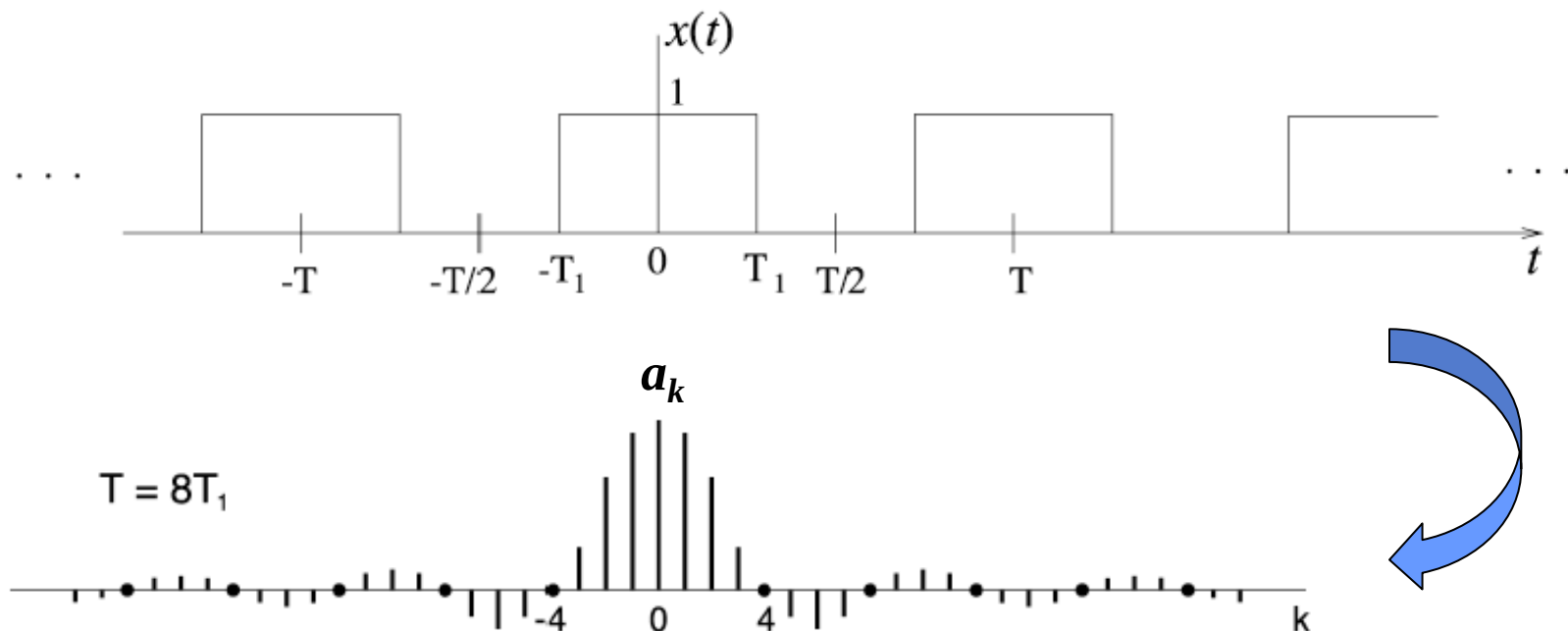
Fourier Transform

- We have shown that Fourier series are useful in analyzing periodic signals, but many (most) signals are aperiodic. Need a more general tool — *Fourier transform*.

Fourier's own derivation of the CT Fourier transform

- $x(t)$ — an aperiodic signal
 - view it as the limit of a periodic signal as $T \rightarrow \infty$



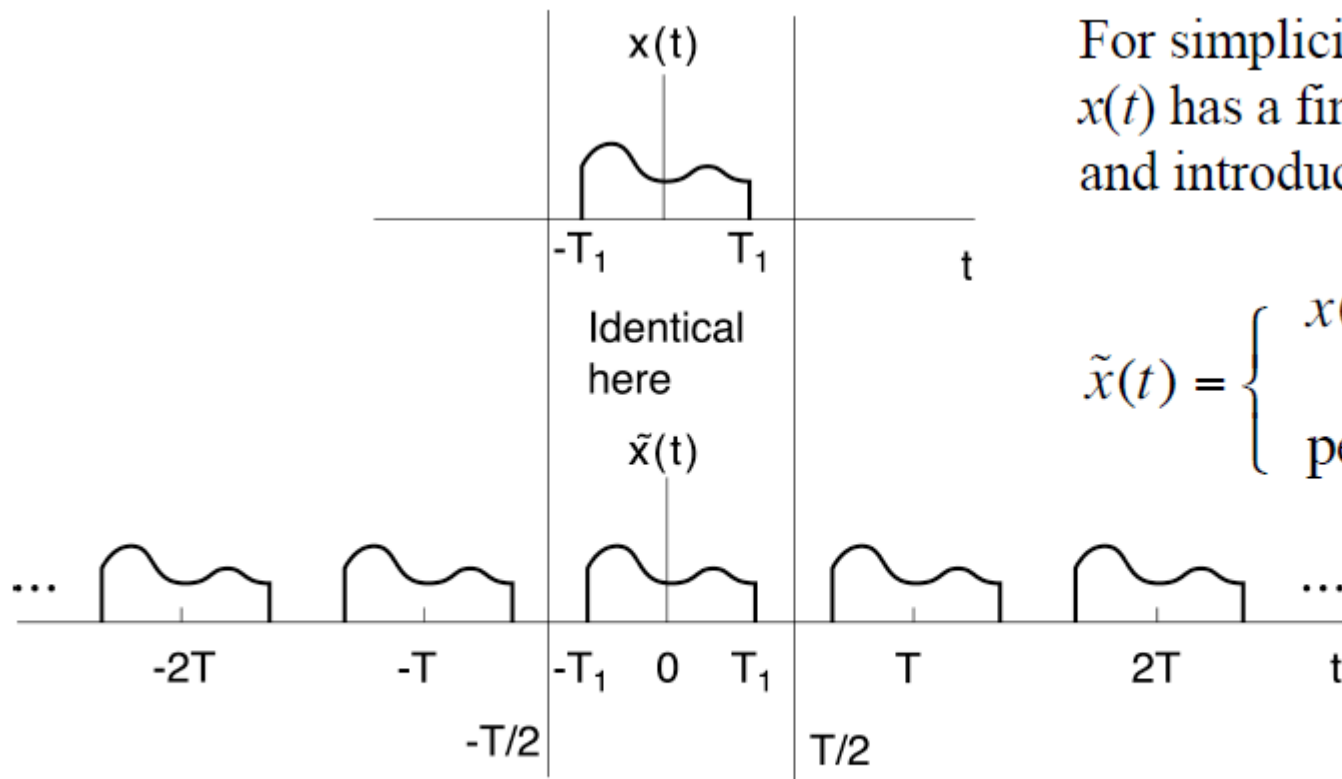


- The harmonic components are spaced $\omega_0 = 2\pi/T$ apart, as $T \rightarrow \infty$, and $\omega_0 \rightarrow 0$, then $\omega = k\omega_0$ becomes continuous



Fourier series $\longrightarrow$ Fourier integral

So, on the derivation of FT ...



As $T \rightarrow \infty$, $x(t) = \tilde{x}(t)$ for all t

Derivation (cont.): Analysis equation

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \quad \omega_o = \frac{2\pi}{T}$$

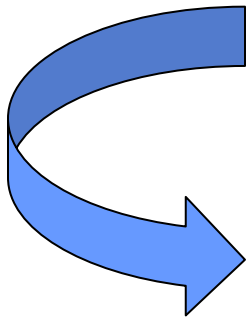
$$a_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \tilde{x}(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-jk\omega_0 t} dt$$

?



$\tilde{x}(t) = x(t)$ in this interval

$$= \frac{1}{T} \int_{-\infty}^{+\infty} x(t) e^{-jk\omega_0 t} dt \quad (1)$$



If we define

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

then, Eq. (1) $\Rightarrow$

$$a_k = \frac{1}{T} X(jk\omega_0) = \frac{1}{T} X(j\omega) |_{\omega=k\omega_0}$$

Derivation (cont.): Synthesis equation

Thus, for $-\frac{T}{2} < t < \frac{T}{2}$

$$x(t) = \tilde{x}(t) = \sum_{k=-\infty}^{+\infty} \underbrace{\frac{1}{T} X(jk\omega_0)}_{a_k} e^{jk\omega_0 t}$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} \omega_0 X(jk\omega_0) e^{jk\omega_0 t}$$

$\Downarrow$

As $T \rightarrow \infty$, $\omega_0 \rightarrow 0$, $\sum \omega_0 \rightarrow \int d\omega$, and $k\omega_0 = \omega$, we get the CT FT pair

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \quad \text{Synthesis equation}$$

— "sum" of $e^{j\omega t}$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \quad \text{Analysis equation}$$

The CT Fourier Transform Pair

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt \quad \text{— } FT$$

Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega \quad \text{— Inverse } FT$$

Inverse Fourier Transform

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$\mathcal{F}(x(t)) = X(j\omega)$$

$$x(t) = \mathcal{F}^{-1}(X(j\omega))$$

CT Fourier Transform Pair

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega$$

CT Fourier Series Pair

$$a_k = \frac{1}{T} \int_T x(t)e^{-jk\omega_0 t} dt$$

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

Harmonically related

For what kinds of signals can we do FT?

It works also even if $x(t)$ is infinite duration, but satisfies:

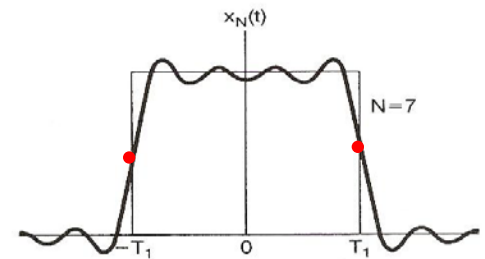
a) Finite energy $\int_{-\infty}^{+\infty} |x(t)|^2 dt < \infty$

In this case, there is *zero* energy in the error

$$e(t) = x(t) - \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad \text{Then} \quad \int_{-\infty}^{\infty} |e(t)|^2 dt = 0$$

b) Dirichlet conditions

- 1) absolutely integrable $\int_{-\infty}^{\infty} |x(t)| dt < \infty$
- 2) finite number of maxima and minima within any finite interval
- 3) finite number of discontinuities with finite values within any finite interval
 - (i) $\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = x(t)$ at points of continuity
 - (ii) $\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega =$ midpoint at discontinuity
 - (iii) Gibb' s phenomenon

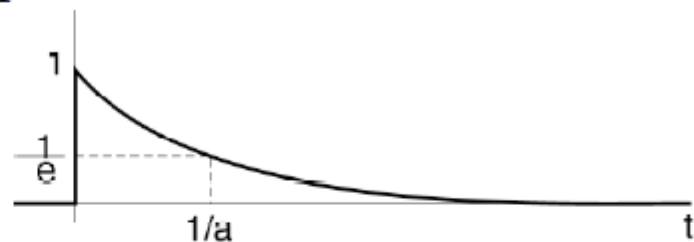


c) By allowing impulses in $x(t)$ or in $X(j\omega)$, we can represent even *more* signals

For example: consider FT for *periodic* signals

Example 4.1 Exponential function

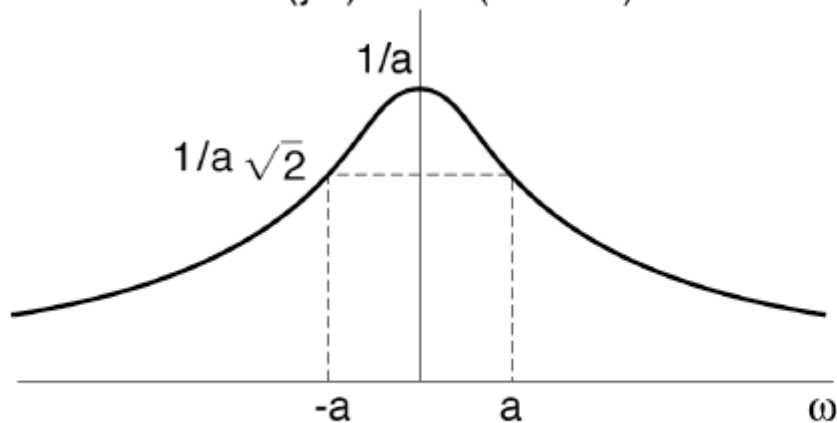
$$x(t) = e^{-at}u(t), \quad a > 0$$



$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt = \int_0^{+\infty} \underbrace{e^{-at}e^{-j\omega t}}_{e^{-(a+j\omega)t}} dt$$

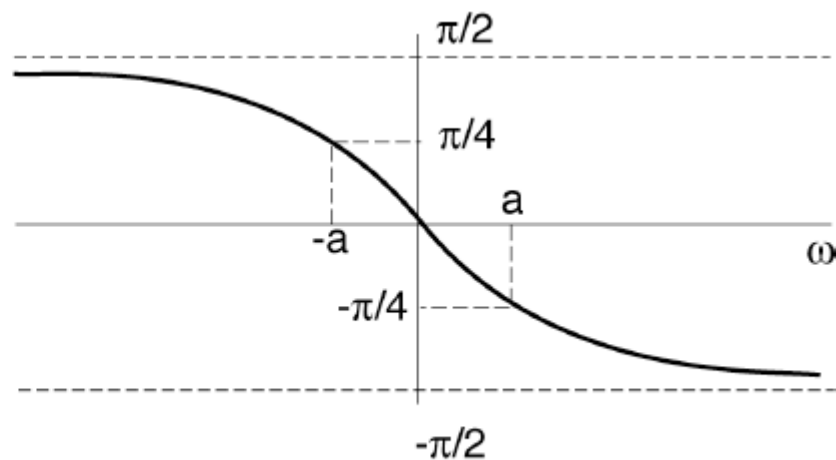
$$= -\left(\frac{1}{a+j\omega}\right)e^{-(a+j\omega)t} \Big|_0^{\infty} = \frac{1}{a+j\omega}$$

$$|X(j\omega)| = 1/(a^2 + \omega^2)^{1/2}$$



Even

$$\angle X(j\omega) = -\tan^{-1}(\omega/a)$$



Odd

Example 4.3 Impulse function

$$(a) \quad x(t) = \delta(t)$$

$$X(j\omega) = \int_{-\infty}^{+\infty} \delta(t) e^{-j\omega t} dt = ?$$

$$\Downarrow$$

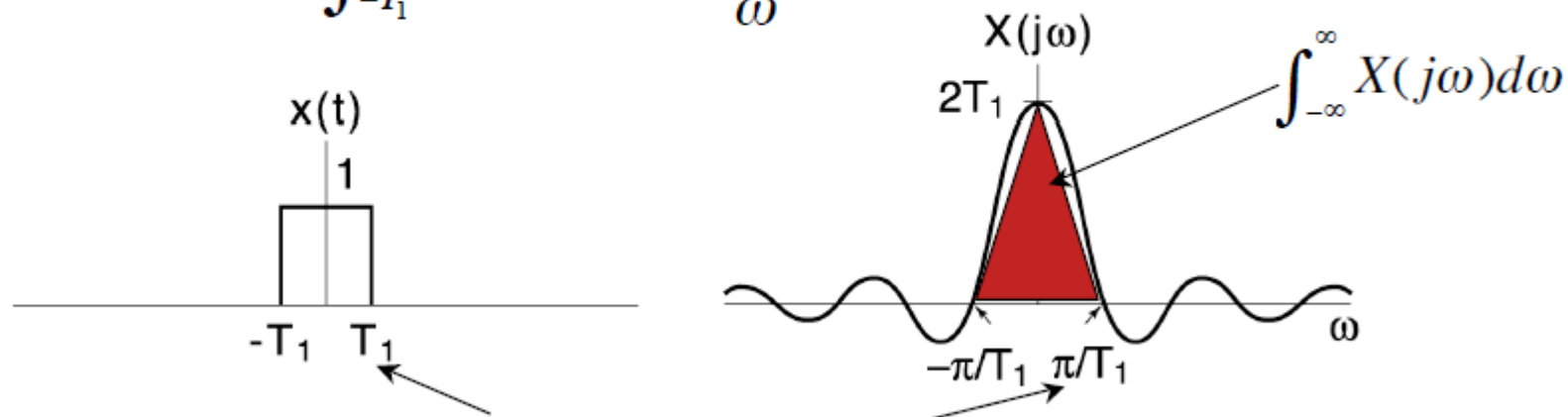
$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{j\omega t} d\omega \quad \text{— Synthesis equation for } \delta(t)$$

$$(b) \quad x(t) = \delta(t - t_0)$$

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{+\infty} \delta(t - t_0) e^{-j\omega t} dt \\ &= e^{-j\omega t_0} \quad \text{— Linear phase shift in } \omega \end{aligned}$$

Example 4.4 A square pulse in the time-domain

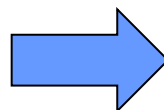
$$X(j\omega) = \int_{-T_1}^{T_1} e^{-j\omega t} dt = \frac{2 \sin \omega T_1}{\omega}$$



Note the inverse relation between the two widths $\Rightarrow$ Uncertainty principle

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$



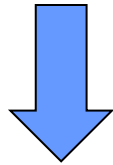
Useful facts about CTFT's

$$X(0) = \int_{-\infty}^{+\infty} x(t) dt$$

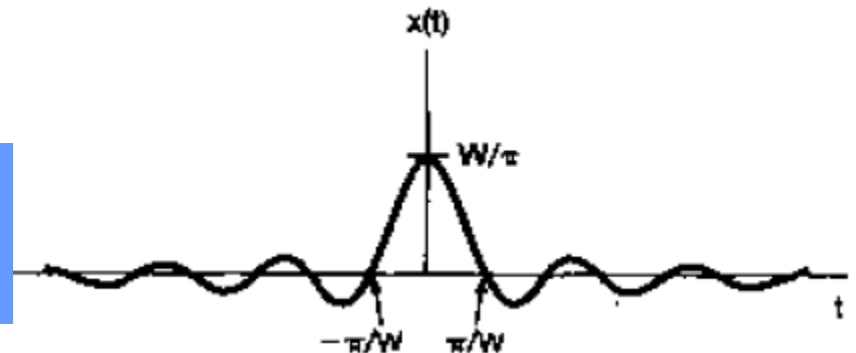
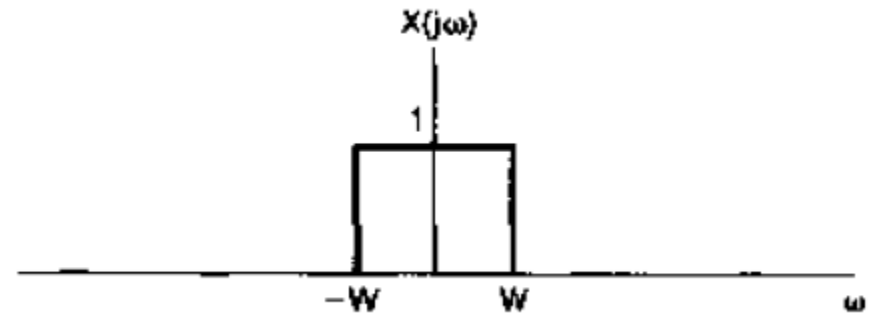
$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) d\omega$$

Example 4.5 A square pulse in the frequency domain

$$X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$



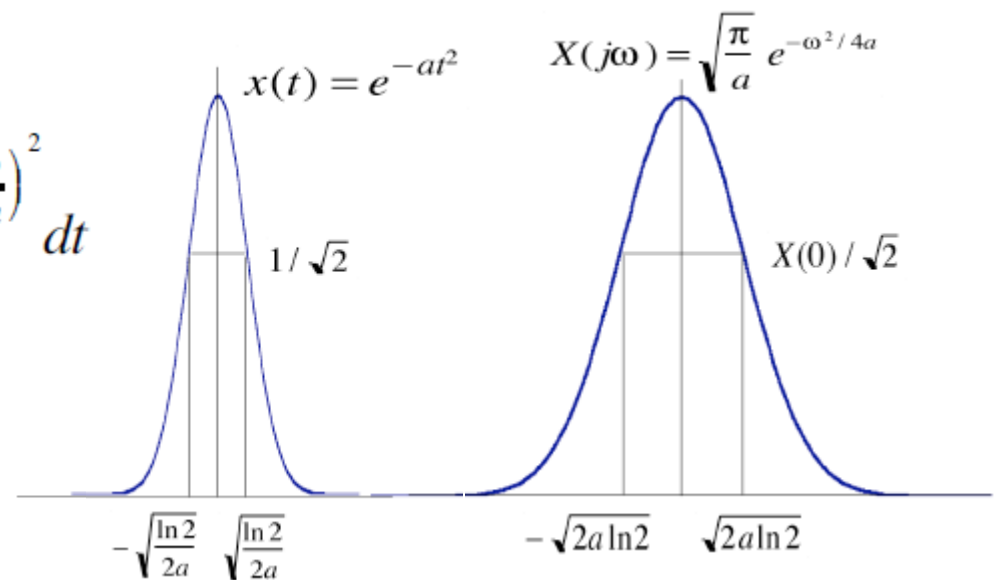
$$x(t) = \frac{1}{2\pi} \int_{-W}^W e^{j\omega t} d\omega = \frac{\sin(Wt)}{\pi t}$$



How about $X(j\omega) = \delta(\omega)$?

Example #4: $x(t) = e^{-at^2}$ — A Gaussian, important in probability, optics, etc.

$$\begin{aligned}
 X(j\omega) &= \int_{-\infty}^{\infty} e^{-at^2} e^{-j\omega t} dt \\
 &= \int_{-\infty}^{\infty} e^{-a \left[t^2 + j\frac{\omega}{a}t + \left(\frac{j\omega}{2a}\right)^2 \right] + a \left(\frac{j\omega}{2a}\right)^2} dt \\
 &= \underbrace{\left[\int_{-\infty}^{\infty} e^{-a \left(t + \frac{j\omega}{2a} \right)^2} dt \right]}_{\sqrt{\pi}/a} \cdot e^{-\frac{\omega^2}{4a}} \\
 &= \sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}
 \end{aligned}$$



(pulse width in t) $\times$ (pulse width in ω)
= a constant

Also a Gaussian!

Uncertainty Principle! Cannot make
both Δt and $\Delta \omega$ arbitrarily small.

CT Fourier Transforms of **Periodic** Signals

Suppose

$$X(j\omega) = \delta(\omega - \omega_0)$$

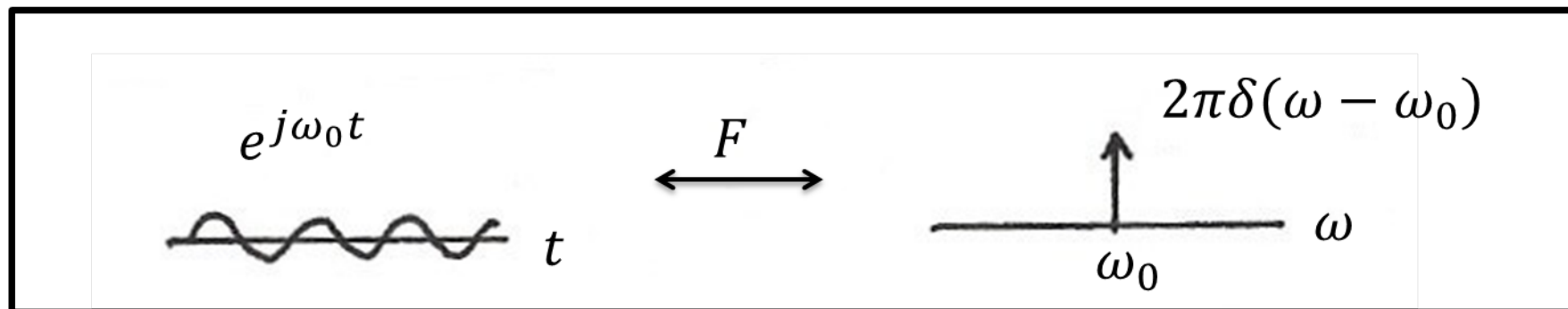
$\Downarrow$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t} \quad \text{— periodic in } t \text{ with frequency } \omega_0$$

That is

$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

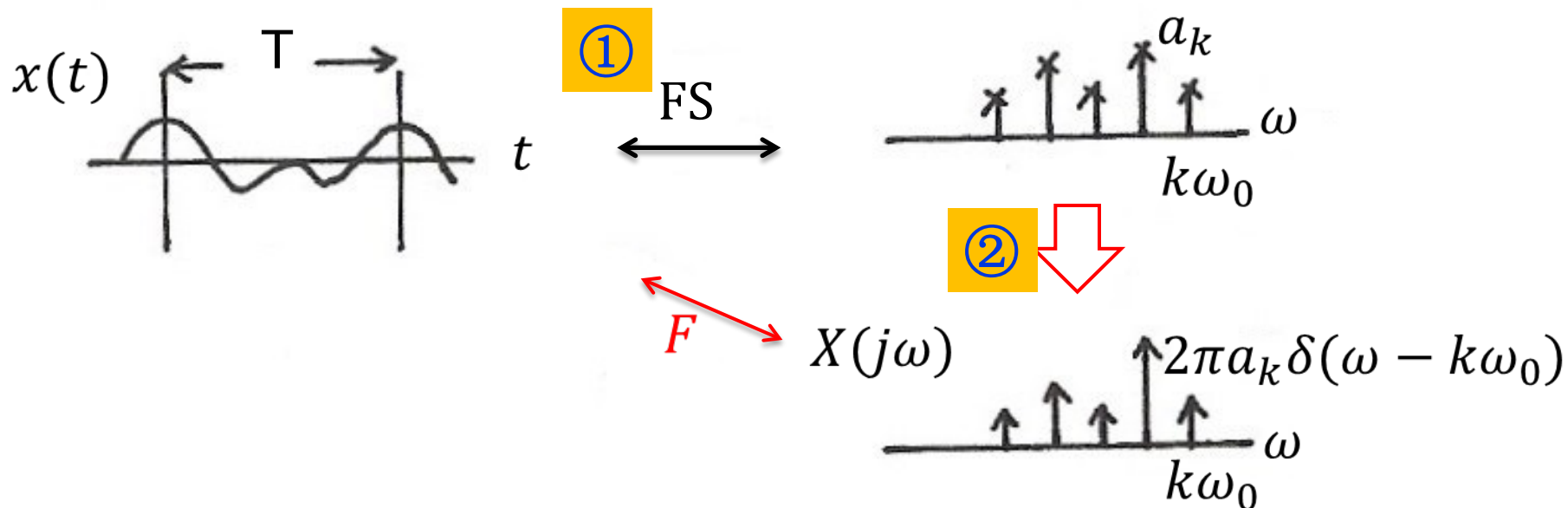
— All the energy is concentrated in one frequency — ω_0



Fourier Transform for Periodic Signals – Unified Framework

More generally, if $x(t) = x(t+T)$, then

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \longleftrightarrow X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0) \quad \text{Discrete spectra}$$

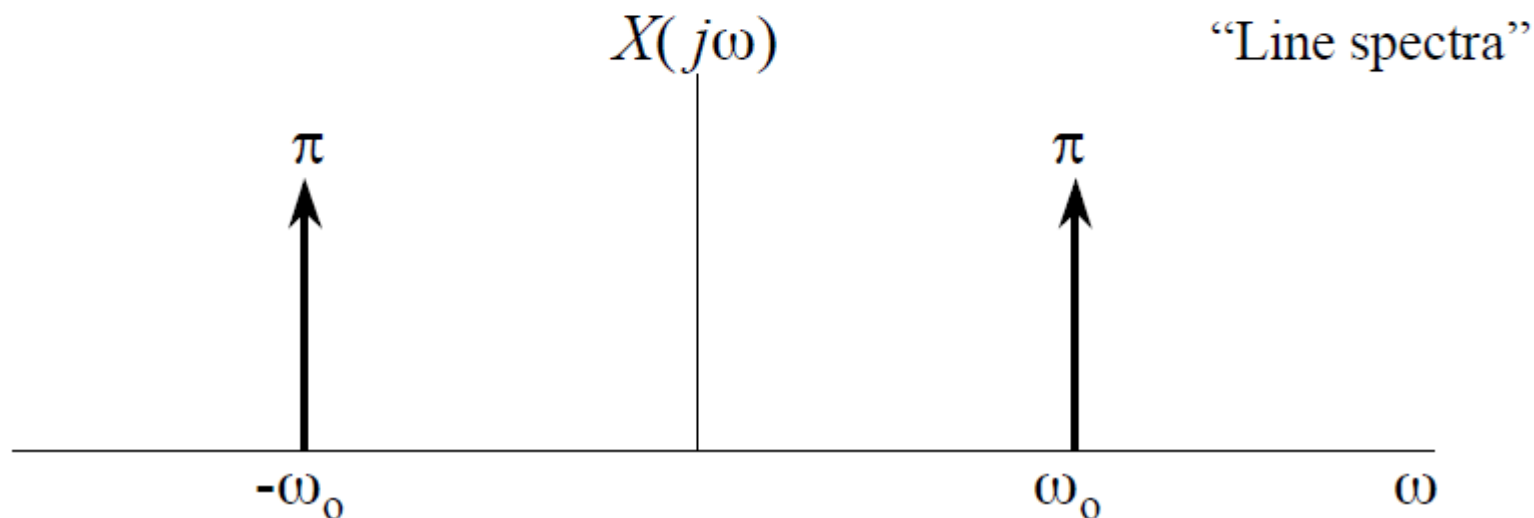


Example 4.7

$$x(t) = \cos \omega_0 t = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}$$



$$X(j\omega) = \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$$



How about $\sin \omega_0 t$?

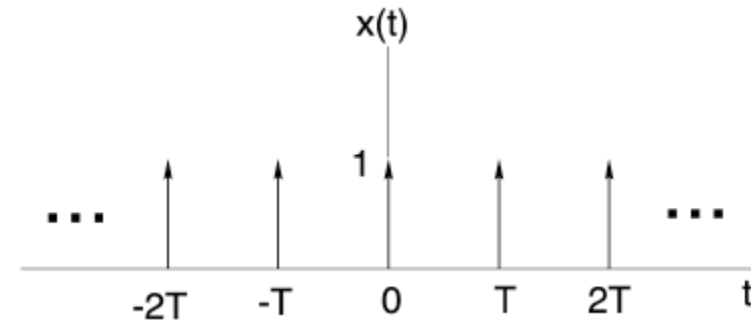
Example 4.8

$$x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad \text{— sampling function}$$

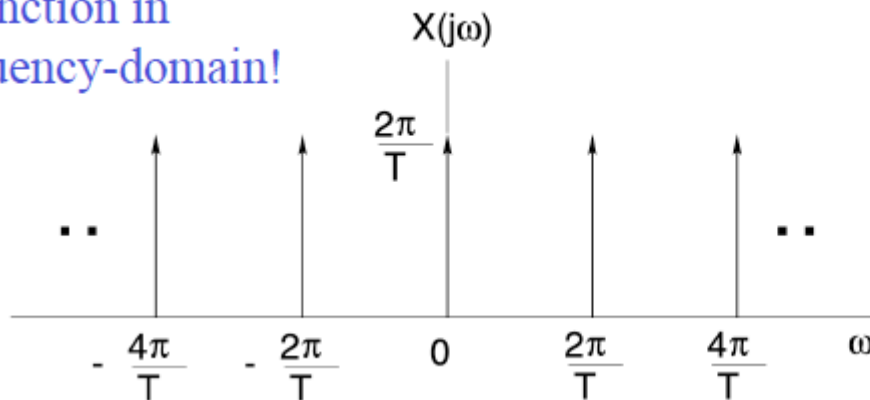
$$x(t) \xleftrightarrow{\text{FS}} a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_o t} dt = \frac{1}{T}$$

$$\Downarrow x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} = \sum_{k=-\infty}^{+\infty} \frac{1}{T} e^{jk\omega_o t}$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2\pi}{T} \delta\left(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_o}\right)$$



Same function in
the frequency-domain!



Note in this case, periodic
in both time domain (with
a period T) and frequency
domain (with a period
 $2\pi/T$)

Properties of the CT Fourier Transform

1) Linearity

$$x(t) \xleftrightarrow{F} X(j\omega), y(t) \xleftrightarrow{F} Y(j\omega)$$

$$ax(t) + by(t) \xleftrightarrow{F} aX(j\omega) + bY(j\omega)$$

2) Time Shifting

$$x(t - t_0) \longleftrightarrow e^{-j\omega t_0} X(j\omega)$$

Proof:
$$\int_{-\infty}^{\infty} x(\underbrace{t - t_0}_{t'}) e^{-j\omega t} dt = e^{-j\omega t_0} \underbrace{\int_{-\infty}^{\infty} x(t') e^{-j\omega t'} dt'}_{X(j\omega)}$$

FT magnitude unchanged

$$|e^{-j\omega_0 t} X(j\omega)| = |X(j\omega)|$$

Linear change in FT phase

$$\angle(e^{-j\omega_0 t} X(j\omega)) = \angle X(j\omega) - \omega t_0$$

CTFT Properties (cont.)

3) Conjugation & Conjugate Symmetry

- Conjugation

$$x^*(t) \xleftrightarrow{F} X^*(-j\omega)$$

- Conjugate Symmetry

$$x(t) \text{ real} \longleftrightarrow X(-j\omega) = X^*(j\omega)$$

$$|X(-j\omega)| = |X(j\omega)| \quad \text{Even} \quad \text{Or} \quad \text{Re}\{X(-j\omega)\} = \text{Re}\{X(j\omega)\} \quad \text{Even}$$

$$\angle X(-j\omega) = -\angle X(j\omega) \quad \text{Odd} \quad \text{Im}\{X(-j\omega)\} = -\text{Im}\{X(j\omega)\} \quad \text{Odd}$$

When $x(t)$ is real (all the physically measurable signals are *real*), the negative frequency components do *not* carry any additional information from the positive frequency components. $\omega \geq 0$ will be sufficient.

CT Fourier Series Property

- Conjugate Symmetry

$$x(t) \text{ real} \Rightarrow a_{-k} = a_k^*$$

Proof:

$$a_{-k} = \frac{1}{T} \int_T x(t) e^{jk\omega_0 t} dt = \left[\frac{1}{T} \int_T x^*(t) e^{-jk\omega_0 t} dt \right]^* = a_k^*$$

$$\therefore a_k = \text{Re}\{a_k\} + j \text{Im}\{a_k\}$$

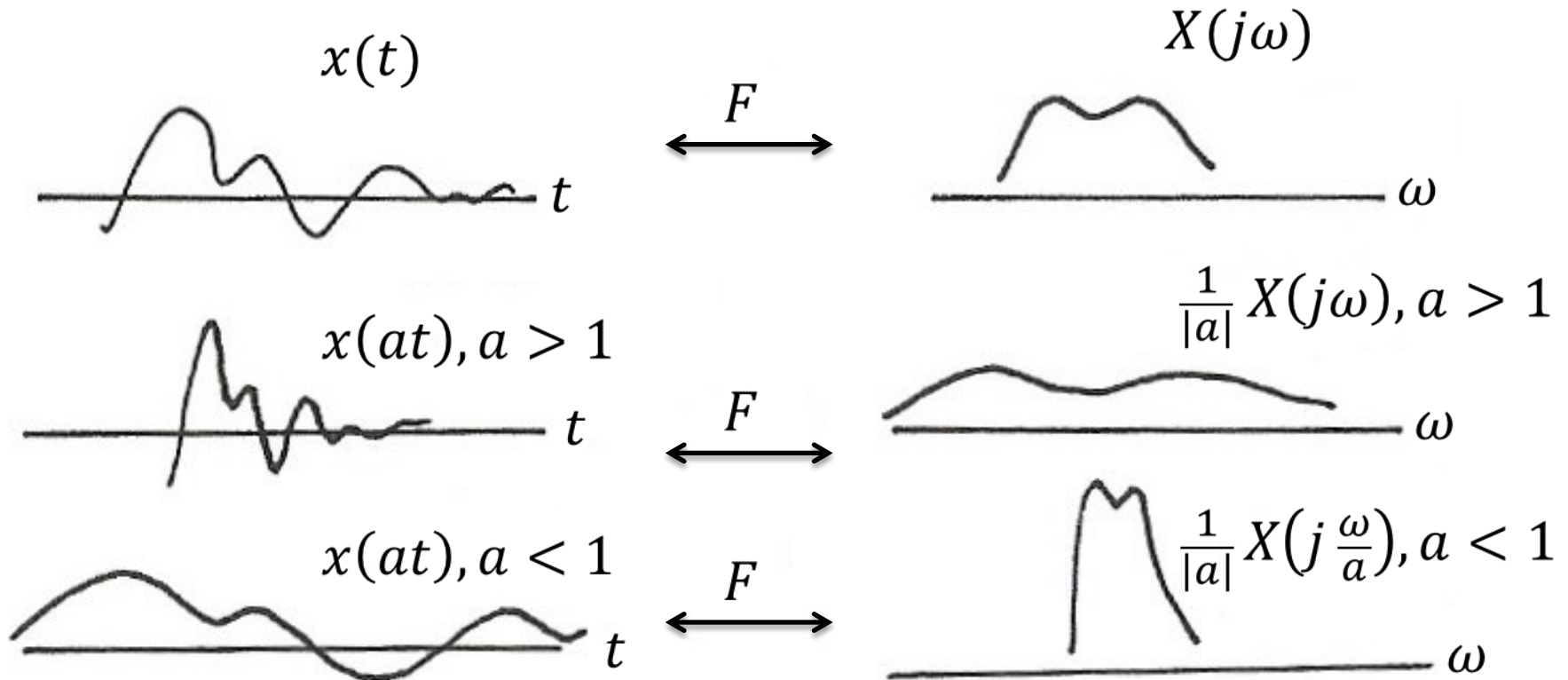
$$\therefore \boxed{\text{Re}\{a_{-k}\} + j \text{Im}\{a_{-k}\}} = \boxed{\text{Re}\{a_k\} - j \text{Im}\{a_k\}}$$

$$\therefore \text{Re}\{a_k\} \text{ is even, } \text{Im}\{a_k\} \text{ is odd}$$

CTFT Properties (cont.)

4) Time/Frequency Scaling $x(at) \longleftrightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$

*E.g. $a > 1 \rightarrow at > t$
compressed in time $\leftrightarrow$
stretched in frequency*



CTFT Properties (cont.)

4) Time/Frequency Scaling $x(at) \longleftrightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$

$\Downarrow a = -1$

$x(-t) \longleftrightarrow X(-j\omega)$

*E.g. $a > 1 \rightarrow at > t$
compressed in time $\leftrightarrow$
stretched in frequency*

Time reversal

$\Downarrow$

a) $x(t)$ real and even $x(t) = x(-t) = x^*(t)$
 $\Rightarrow X(j\omega) = X(-j\omega) = X^*(j\omega)$ — Real & even

b) $x(t)$ real and odd $x(t) = -x(-t) = x^*(t)$
 $\Rightarrow X(j\omega) = -X(-j\omega) = -X^*(j\omega)$ — Purely imaginary
 & odd

c) $X(j\omega) = \text{Re}\{X(j\omega)\} + j\text{Im}\{X(j\omega)\}$

For real $x(t) = \text{Ev}\{x(t)\} + \text{Od}\{x(t)\}$

inverse
relationship
between signal
“width” in
time/frequency
domains

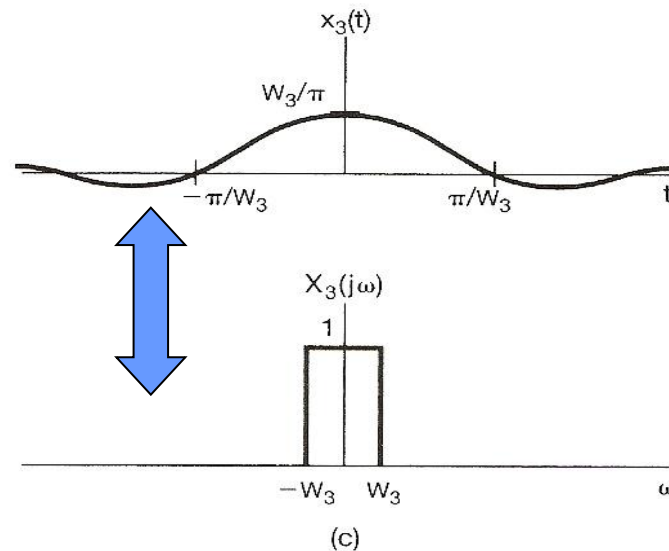
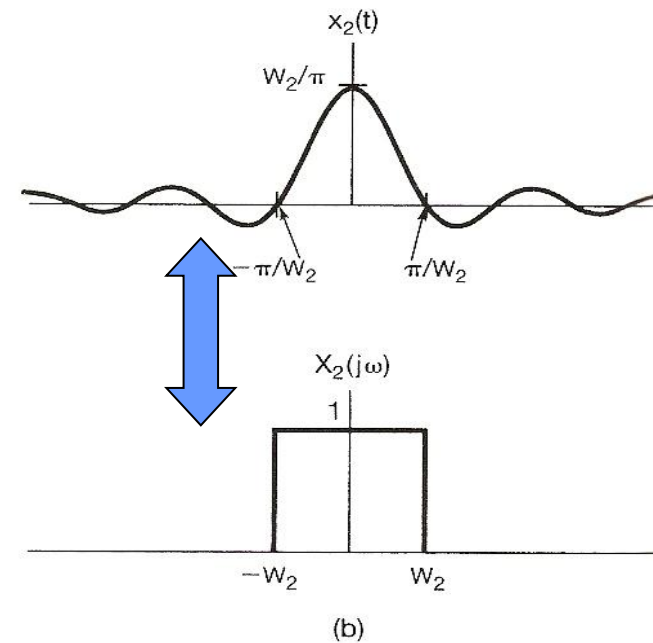
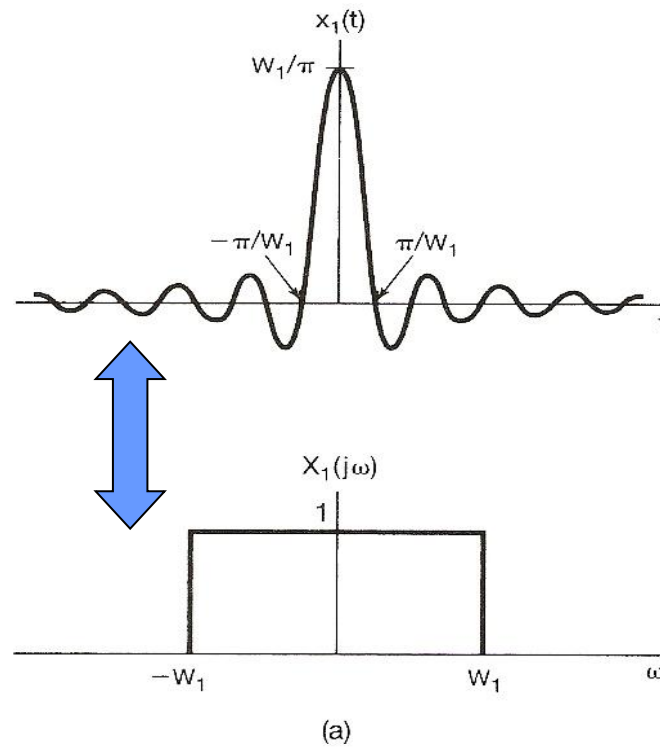


Figure 4.11 Fourier transform pair of Figure 4.9 for several different values of W .

CTFT Properties (cont.)

5) Differentiation/Integration

$$\frac{dx(t)}{dt} \xleftrightarrow{F} j\omega X(j\omega)$$

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{F} \frac{1}{j\omega} X(j\omega) + \pi X(j0) \delta(\omega)$$

$\uparrow$
 DC term

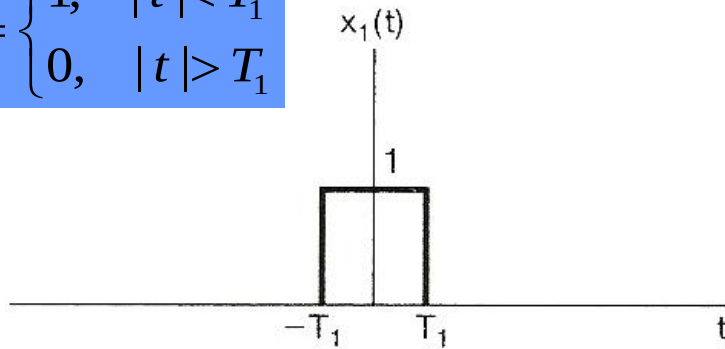
Example:

What is the Fourier transform for unit step function $u(t)$?

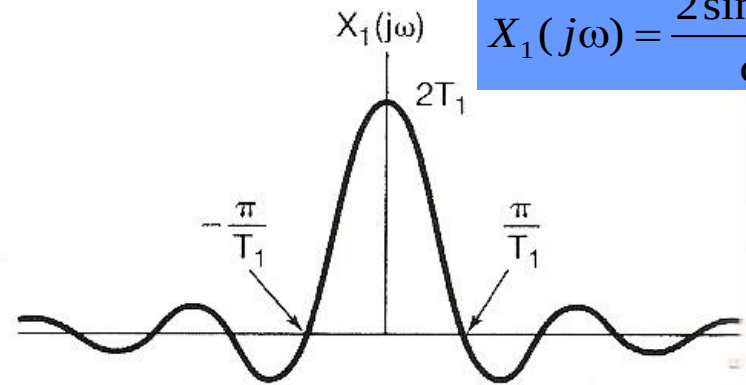
CTFT Properties (cont.)

6) Duality

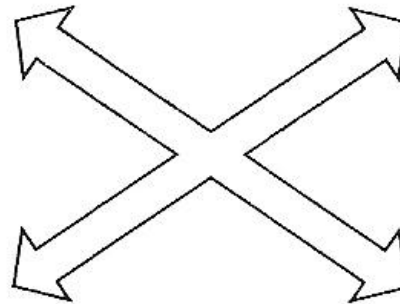
$$x_1(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$$



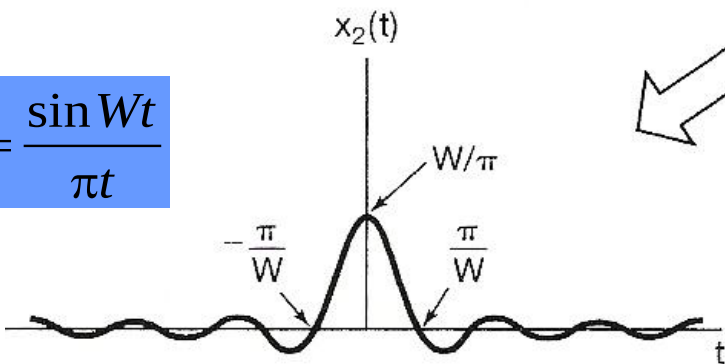
$\mathcal{F}$



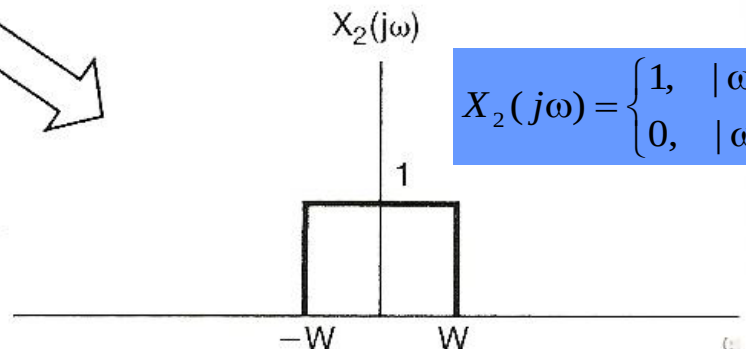
$$X_1(j\omega) = \frac{2 \sin \omega T_1}{\omega}$$



$$x_2(t) = \frac{\sin Wt}{\pi t}$$



$\mathcal{F}$

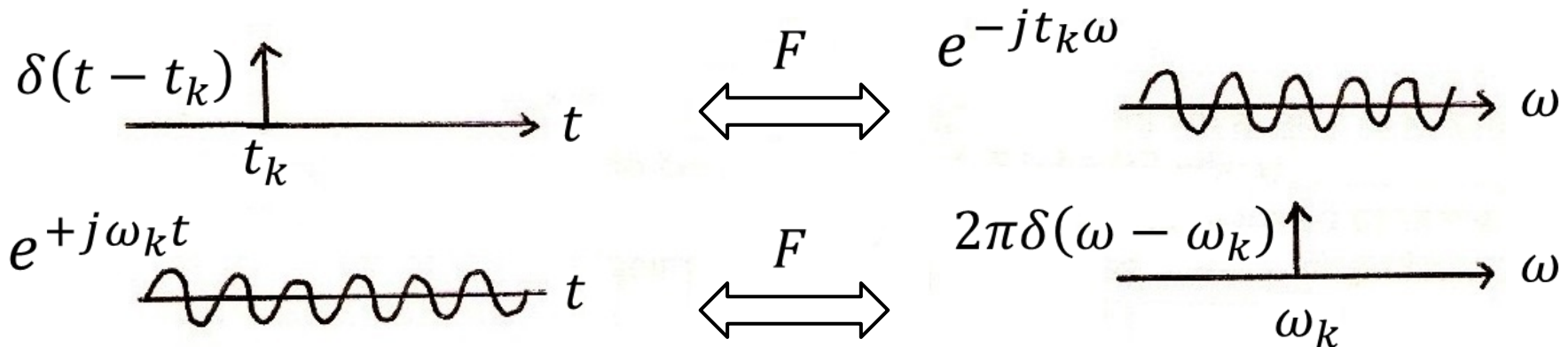


$$X_2(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$

- Time/frequency domains are kind of “symmetric”.
- If there are characteristics of a function of time that have implications with regard to the Fourier transform, then the same characteristics associated with a function of frequency will have **dual** implications in the time domain.

Example:

$$\{\delta(t - t_k), -\infty < t_k < \infty\} \quad \{2\pi\delta(\omega - \omega_k), -\infty < \omega_k < \infty\}$$



CTFT Properties (cont.)

7) Parseval's Relation

$$\underbrace{\int_{-\infty}^{+\infty} |x(t)|^2 dt}_{\text{Total energy in the time-domain}} = \underbrace{\frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega}_{\text{Total energy in the frequency-domain}}$$

Total energy in the
time-domain

Total energy in the
frequency-domain

$$\frac{1}{2\pi} |X(j\omega)|^2$$

— spectral density

Table 4.2

Basic Fourier Transform Pairs

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0$, otherwise
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1$, $a_k = 0$, $k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave		
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$ for all k
$x(t) \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi \delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

Summary

- **Understand CT Fourier transform**
 - ◆ **Synthesis and analysis equations**
 - ◆ **Difference with CT Fourier series**
 - ◆ **Fourier transform for periodic signal**
 - ◆ **Properties of CT Fourier transform**